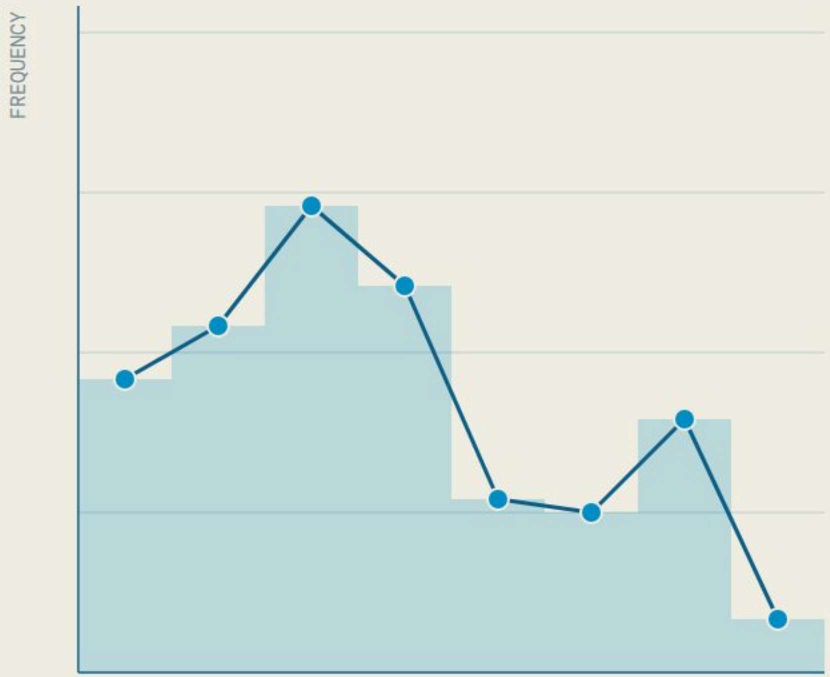
CHAPTER TWO

Describing data: *tables and charts*



Frequency tables, frequency distributions, and graphic presentation: how to turn a list of numbers into a picture you can read.

3
LEARNING OBJECTIVES

36
SLIDES

25
PRACTICE POOLS

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Three learning objectives

2.1

QUALITATIVE DATA

Organize and display qualitative data using **frequency** and **relative frequency tables**, **bar charts**, and **pie charts**.

2.2

QUANTITATIVE DATA

Organize and display quantitative data using **frequency** and **relative frequency distributions** and **histograms**.

2.3

MORE CHARTS

Visualize quantitative data using **frequency polygons**, **cumulative frequency distributions** and **dot plots**.

One story runs through the chapter: a retail store sold **160 units** last month. First we look at *what* was sold (a qualitative variable), then at the *profit* on each unit (a quantitative variable). Two practice games and 25 practice pools support this chapter.

2.1

CONTENTS · 8 SLIDES

LEARNING OBJECTIVE

Frequency tables, *bar charts* and pie charts

A qualitative variable puts each unit into a category. The first job of statistics is to count how many are in each category, and then show it.

What is a frequency table?

LO 2.1 · Frequency tables, bar charts and pie charts

DEFINITION

A **frequency table** is a grouping of qualitative data into **mutually exclusive** and **collectively exhaustive** classes, showing the number of observations in each class.

01 Mutually exclusive

Each observation fits in only **one** class. A student cannot be counted in both Business and Medicine.

02 Collectively exhaustive

Every observation fits **somewhere** in the table. Every student's major is included. If needed, add a class “Other”.

03 Frequency

The **count** of observations in a class. The frequencies add up to the total number of observations.

Table 1 Frequency table of the majors of 50 students in one class

MAJOR	NUMBER OF STUDENTS
Business	15
Engineering	12
Medicine	18
Arts	5
Total	**50**

Each student has exactly one major (mutually exclusive) and every student appears in one row (collectively exhaustive). 15 + 12 + 18 + 5 = 50.

How to build a frequency table

LO 2.1 · Frequency tables, bar charts and pie charts

A retail store in Riyadh sold **160 units** last month. For every unit the store recorded the **product category**, a qualitative (nominal) variable.

01 **Sort**

Put every unit into its category: Electronics, Apparel, Home Goods or Books.

02 **Count**

Count how many units are in each category.

03 **Report**

Write the count as the **frequency** of the category, and add the total.

CHECK
66 + 47 + 27 + 20 = 160. The frequencies add up to the total.

Table 2 Frequency table of the 160 units sold last month, by product category

PRODUCT CATEGORY	NUMBER OF UNITS SOLD
Electronics	66
Apparel	47
Home Goods	27
Books	20
Total	**160**

If 10 more Books were sold, what would the new total be?

170. The table changes in two places: Books and Total.

Relative *and* percent frequency

LO 2.1 · Frequency tables, bar charts and pie charts

RELATIVE FREQUENCY

$$\text{relative frequency} = \frac{\text{frequency of the class}}{\text{total number of observations}}$$

It shows what **fraction** of the total each category represents. Relative frequencies add up to 1.

PERCENT FREQUENCY

$$\text{percent} = \text{relative frequency} \times 100$$

The same information as a percentage. Percents add up to 100%.

READ IT IN WORDS
Electronics account for 41.25% of all units sold.

Table 3 Relative and percent frequency of the units sold, by product category

CATEGORY	FREQUENCY	CALCULATION	RELATIVE FREQUENCY	PERCENT
Electronics	66	66 / 160	0.4125	41.25%
Apparel	47	47 / 160	0.29375	29.375%
Home Goods	27	27 / 160	0.16875	16.875%
Books	20	20 / 160	0.125	12.5%
Total	**160**	160 / 160	**1.000**	**100%**

If Books were 50% of all units, how many units would that be?

0.50 × 160 = 80 units.

Bar charts

LO 2.1 · Frequency tables, bar charts and pie charts

Figure 1 Bar chart of the units sold by product category

HOW TO READ IT

Categories on the horizontal axis, frequencies on the vertical axis. The **height** of each bar is the frequency of that category.

The bars are **separated by gaps** because the categories are separate things; there is nothing “between” Apparel and Books.

THE MODE

The category with the tallest bar is the **mode**, the most frequent category. Here: Electronics (66 units).

Bar charts work best for distinct categories: favourite colours, smartphone brands, university majors, payment methods.

The order of the bars

LO 2.1 · Frequency tables, bar charts and pie charts

For a **nominal** variable (categories with no natural order) you may arrange the bars in any order: alphabetical, decreasing frequency, or increasing frequency. The chart is correct in every order, but **ordering by frequency** usually makes the pattern easier to see.

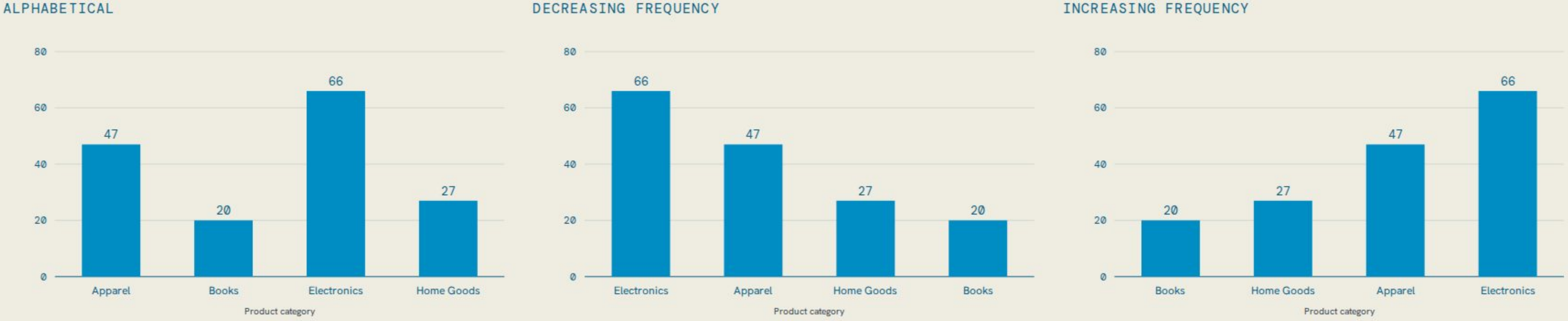


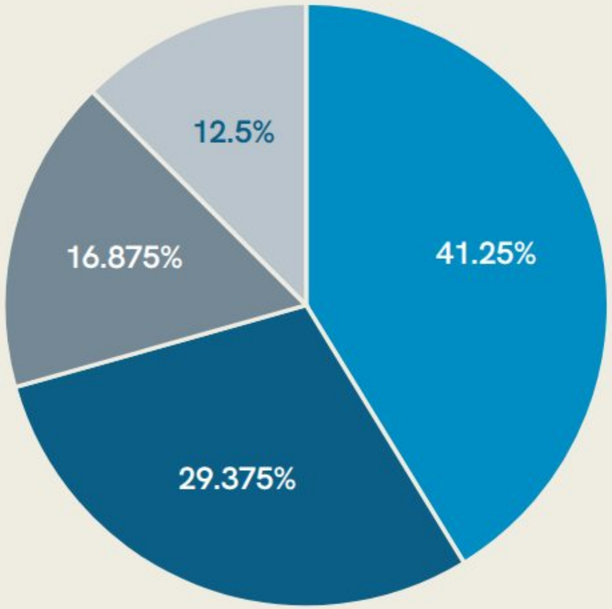
Figure 2 The same bar chart in three orders: alphabetical, decreasing and increasing frequency

For an **ordinal** variable (poor · fair · good · excellent, or 1★ to 5★) keep the natural order of the categories and do not sort by frequency. Question: would the chart look different if we arranged the products alphabetically? Yes: the bars move, but every bar keeps its height.

Pie charts: parts *of a whole*

LO 2.1 · Frequency tables, bar charts and pie charts

SHARE OF UNITS SOLD · 160 = 100%



Angle of each slice = share × 360°
Electronics: 0.4125 × 360° = 148.5°

Electronics
66 of 160 units → 66/160 = 0.4125 →

Apparel
47 of 160 units → 47/160 = 0.2938 →

Home Goods
27 of 160 units → 27/160 = 0.1688 →

Books
20 of 160 units → 20/160 = 0.125 →

Figure 3 Pie chart of the units sold by product category

HOW IT WORKS

The whole circle is **100%** of the observations. Each category is a **slice**; the size of the slice is the relative frequency of the category. Bigger slice = more popular category.

PIE CHARTS WORK BEST WHEN

- There are few categories (about 7 or fewer).
- You want to show how each part relates to the whole.
- The differences between the categories are clear.

The pie chart is drawn from the **relative frequency table**: 66/160 = 41.25% → 148.5° of the circle.

Bar chart *or* pie chart?

LO 2.1 · Frequency tables, bar charts and pie charts

Table 4 Bar chart or pie chart: when to use each

	BAR CHART	PIE CHART
Question it answers	**Which category is bigger?** Compare the heights.	**What share of the whole?** Compare the slices.
Best when	Many categories, or the counts are close to each other.	Few categories and the parts of the whole matter (market share, budget split).
Data needed	Frequencies (or relative frequencies).	Relative frequencies (percentages); the slices must add up to 100%.
Weak point	Does not show the share of the whole directly.	Hard to compare slices of similar size; too many slices become unreadable.
Retail example	Electronics 66 vs. Apparel 47: a difference of 19 units is easy to see.	Electronics is 41% of all units, a bit less than half of the pie.

Both charts show the same frequency table. Both start with the table: **the table comes first, the chart comes second.**

Check your understanding

LO 2.1 · Frequency tables, bar charts and pie charts

Q1 What is the difference between frequency and relative frequency?

Frequency is the count; relative frequency is the count divided by the total (a fraction of the whole).

Q2 35 students prefer studying at home and 65 prefer the library. What is the relative frequency of "at home"? What is the percent frequency?

35/(35 + 65) = 35/100 = 0.35, that is 35%.

Q3 A bank has 5 account types and wants to show which type is most common. Bar chart or pie chart? And if it wants to show each type's share of all accounts?

Bar chart to compare; pie chart to show shares (5 slices is fine).

Q4 A frequency table of nationalities lists Saudi, Egyptian, Indian and Pakistani customers, but 12 customers are from other countries and appear nowhere. Which property is broken?

Collectively exhaustive; add a class "Other".

2.2

CONTENTS · 8 SLIDES

LEARNING OBJECTIVE

Frequency *distributions* and histograms

A quantitative variable gives a number for every unit. 160 numbers are hard to read, so we group them into classes and count. Then we draw the histogram.

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What is a frequency distribution?

LO 2.2 · Frequency distributions and histograms

DEFINITION

A **frequency distribution** organizes quantitative (numerical) data into mutually exclusive and collectively exhaustive **classes**, showing the number of observations in each class.

The idea is the same as the frequency table, but the "categories" are now **intervals of numbers** (for example 0 up to 200, 200 up to 400 …), and we have to decide the intervals ourselves.

WHY GROUP?

160 numbers → 8 classes → a shape you can see.

01 **Decide how many classes**

Use the 2^k rule: the smallest k with $2^k > n$.

02 **Determine the class width**

(maximum − minimum) ÷ k, then round up to a convenient number.

03 **Set the class limits**

Start at or below the minimum; each class is "lower limit up to upper limit".

04 **Count the values in each class**

Tally, then write the frequencies. They must add up to n.

The raw data: *profit* on 160 units

LO 2.2 · Frequency distributions and histograms

The same store also recorded the **profit** (in riyals) on each of the 160 units, a quantitative variable. Profits ranged from a minimum of ⃁ 53 to a maximum of ⃁ 1,491. In this raw form the data is hard to understand. We need to organize it.

Table 5 Raw data: profit on each of the 160 units sold last month (minimum and maximum highlighted)

577	623	739	918	60	554	1,388	408	556	431	934	384	1,347	1,480	1,128	238	1,349	288	1,186	60
416	854	1,214	528	528	1,210	958	739	60	60	237	788	135	734	563	923	823	806	75	609
181	1,363	611	1,351	567	542	228	705	525	474	259	1,039	**1,491**	460	303	741	133	60	1,037	419
635	230	1,305	137	924	1,314	623	1,305	490	1,371	324	653	775	641	338	301	819	564	426	1,308
735	1,346	1,151	939	1,358	662	1,109	72	1,029	762	488	73	179	575	226	357	736	288	102	349
771	1,480	168	138	238	611	220	435	672	408	941	640	218	409	477	344	139	495	364	234
585	1,220	506	60	1,381	766	312	911	730	515	634	708	245	1,249	1,102	1,222	635	518	440	699
569	1,155	177	230	218	1,094	168	1,480	425	584	1,183	785	570	1,249	**53**	155	239	1,187	946	474

Step 1 · How many classes?

LO 2.2 · Frequency distributions and histograms

THE RULE

choose the smallest k such that $2^k > n$

k = number of classes, n = number of observations. Usually between 5 and 15 classes; too few hide the shape, too many show noise.

OUR DATA

$n = 160$:

$2^7 = 128 < 160$, and

$2^8 = 256 > n$; stop

so $k = 8$

With 400 data points, how many classes?

$2^8 = 256 < 400$ and $2^9 = 512 > 400$, so k = **9 classes**.

Table 6 Finding k with the 2^k rule for $n = 160$

k	2^k	$2^k > n = 160$?
1	2	no
2	4	no
3	8	no
4	16	no
5	32	no
6	64	no
7	128	no
8	256	yes, so $k = 8$

Step 2 · Class width

LO 2.2 · Frequency distributions and histograms

THE FORMULA

$$\text{class width} \quad i \geq \frac{\text{maximum value} - \text{minimum value}}{k}$$

$$i \geq \frac{₱\,1{,}491 - ₱\,53}{8} = \frac{₱\,1{,}438}{8} = 179.75$$

179.75 is an awkward number. **Round up** to a convenient width: i = **200**. Rounding *down* would leave the largest value outside the last class. Every class will now cover a range of ₱ 200 of profit.

RULES OF THUMB

- All classes have the **same width**.
- Round up to a “nice” number: 10, 20, 25, 50, 100, 200, 500 ...
- The classes must cover the whole range: from below the minimum to above the maximum.

THINK OF IT AS

Drawers in a filing cabinet: too narrow means too many drawers; too wide means everything gets mixed together.

SECOND EXAMPLE

Exam scores of 30 students, from 45 to 98. Step 1: 2^5 = 32 > 30, so k = 5. Step 2: (98 – 45)/5 = 10.6, round up to **11**. Classes: 45 up to 56, 56 up to 67, 67 up to 78, 78 up to 89, 89 up to 100.

Step 3 · Class limits

LO 2.2 · Frequency distributions and histograms

Start at a convenient number **at or below the minimum** (53 → start at 0) and add the class width to make each new class. Each class has a **lower limit** and an **upper limit**.

THE "UP TO" CONVENTION

"0 up to 200" means: from 0 **up to but not including** 200. A profit of exactly ⊭ 200 goes in the **next** class, "200 up to 400". So the classes are mutually exclusive: no value can be in two classes.

EXAMPLE

A profit of exactly ⊭ 1,000 goes in "1,000 up to 1,200", not in "800 up to 1,000".

Table 7 The eight profit classes and their limits

CLASS	LOWER LIMIT	UPPER LIMIT
0 up to 200	0	200
200 up to 400	200	400
400 up to 600	400	600
600 up to 800	600	800
800 up to 1,000	800	1,000
1,000 up to 1,200	1,000	1,200
1,200 up to 1,400	1,200	1,400
1,400 up to 1,600	1,400	1,600

Step 4 · Count the frequencies

LO 2.2 · Frequency distributions and histograms

Table 8 Frequency distribution of the profit on 160 units

PROFIT (⃁)	FREQUENCY
0 up to 200	22
200 up to 400	26
400 up to 600	35
600 up to 800	29
800 up to 1,000	13
1,000 up to 1,200	12
1,200 up to 1,400	19
1,400 up to 1,600	4
Total	**160**

Go through the 160 values and put a tally mark in the class of each value. Then count the marks. The frequencies must add up to **160**.

WHAT THE TABLE TELLS US

- Most units (35) made a profit of 400 up to 600 riyals.
- Very few units (4) made 1,400 riyals or more.
- The profits are **not** spread evenly: a big group in the middle-low range, then a second smaller bump at 1,200 up to 1,400.

How many units made a profit of at least ⃁ 1,200?

19 + 4 = **23 units**.

Relative frequency distribution

LO 2.2 · Frequency distributions and histograms

Table 9 Relative frequency distribution of the profit on 160 units

PROFIT (⃁)	FREQUENCY	RELATIVE FREQUENCY	PERCENT
0 up to 200	22	22/160 = 0.1375	13.75%
200 up to 400	26	26/160 = 0.1625	16.25%
400 up to 600	35	35/160 = 0.2188	21.88%
600 up to 800	29	29/160 = 0.1812	18.12%
800 up to 1,000	13	13/160 = 0.0813	8.12%
1,000 up to 1,200	12	12/160 = 0.0750	7.50%
1,200 up to 1,400	19	19/160 = 0.1187	11.88%
1,400 up to 1,600	4	4/160 = 0.0250	2.50%
Total	**160**	**1.0000**	**100%**

Exactly as with qualitative data: divide each frequency by the total. The relative frequency distribution shows what **share** of the units falls into each profit range.

READ IT IN WORDS

21.88% of the units made a profit of 400 up to 600 riyals.

What percentage of units had a profit of less than ⃁ 1,400?

100% − 2.50% = **97.5%** (or add the first seven percents).

Relative frequencies let us compare groups of **different sizes**: this month's 160 units with next month's 240 units.

Histograms

LO 2.2 · Frequency distributions and histograms

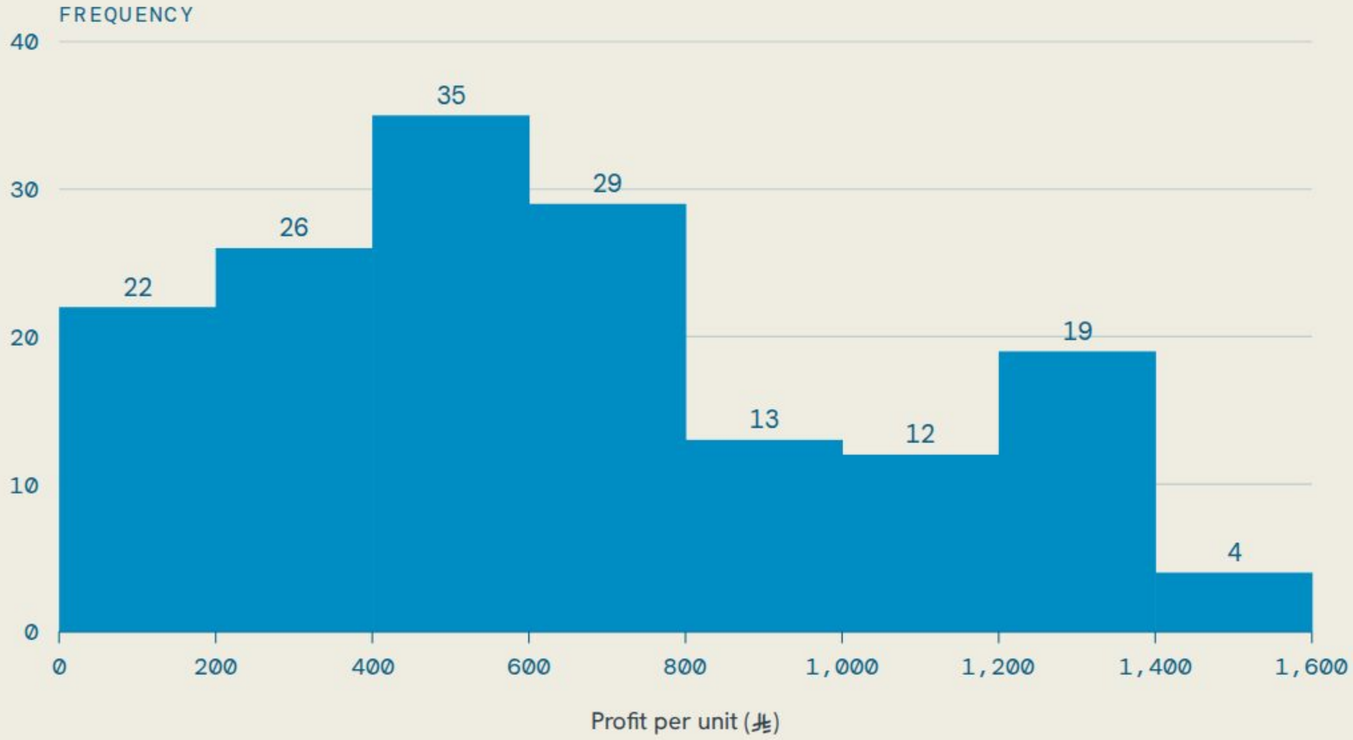

Figure 4 Histogram of the profit on 160 units

KEY FEATURES

- Classes on the horizontal axis.
- Frequencies on the vertical axis.
- The bars **touch** each other, no gaps.
- The height of each bar is the frequency of the class.

WHY NO GAPS?

The data is numerical and continuous: a profit of 199.5 or 200.5 is possible. A gap would suggest that no values can exist between the classes.

What we learn: most units are in 400 up to 600; very few above 1,400; the shape is **not symmetric**: a long tail to the right with a small second bump.

Histogram *vs.* bar chart

LO 2.2 · Frequency distributions and histograms

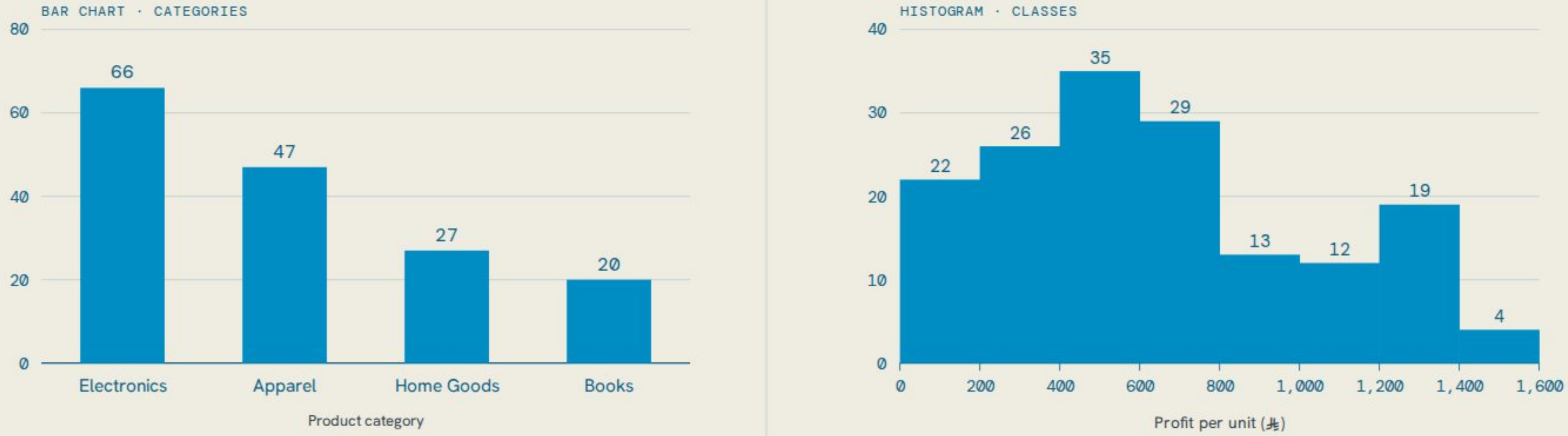

Figure 5 A bar chart of categories (left) and a histogram of numerical classes (right)

Table 10 Bar chart and histogram compared

	BAR CHART	HISTOGRAM
Variable	Qualitative: categories	Quantitative: numbers grouped into classes
Bars	Separated by gaps; any order for nominal data	Touch each other; in numerical order
Horizontal axis	Category names	A number line with the class limits

Check your understanding

LO 2.2 · Frequency distributions and histograms

Q1 **Why do we group numerical data into classes?**

To turn a long list into a manageable table that shows the pattern (where most values are, the spread, the shape).

Q2 **Exam scores of 30 students range from 45 to 98. How many classes does the 2^k rule suggest, and what class width would you use?**

$2^4 = 16 < 30 < 32 = 2^5$, so $k = 5$ classes; width $i \geq (98 - 45)/5 = 10.6$, round up to 11 (classes 45 up to 56, 56 up to 67, ..., 89 up to 100).

Q3 **In the retail data, a unit made a profit of exactly ⃁ 600. Which class does it belong to?**

"600 up to 800", because the upper limit is not included.

Q4 **What is the difference between a histogram and a bar chart?**

Histogram: numerical classes, bars touch. Bar chart: categories, bars have gaps.

2.3

LEARNING OBJECTIVE

Frequency *polygons*, cumulative distributions and dot plots

Two more charts of numerical data: the polygon shows the shape with one line (good for comparing), and the dot plot shows every single value. In between, the cumulative distribution answers “how many are below...?”

CONTENTS · 11 SLIDES

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Frequency polygons

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

HOW TO DRAW ONE

1. Find the **midpoint** of each class: halfway between the lower and the upper limit.
2. Plot a point at (midpoint, frequency).
3. Connect the points with straight lines.
4. Close the polygon: add a point with frequency 0 one class before the first and one class after the last.

$$\text{midpoint} = \frac{\text{lower limit} + \text{upper limit}}{2}$$

$$\text{for example:} \quad \frac{200 + 400}{2} = 300$$

What are the coordinates (midpoint, frequency) of the class with the highest frequency?

(500, 35).

Table 11 Class midpoints and the points of the frequency polygon

PROFIT (⃁)	MIDPOINT	FREQUENCY	POINT
0 up to 200	100	22	(100, 22)
200 up to 400	300	26	(300, 26)
400 up to 600	500	35	(500, 35)
600 up to 800	700	29	(700, 29)
800 up to 1,000	900	13	(900, 13)
1,000 up to 1,200	1,100	12	(1,100, 12)
1,200 up to 1,400	1,300	19	(1,300, 19)
1,400 up to 1,600	1,500	4	(1,500, 4)

The polygon over the histogram

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

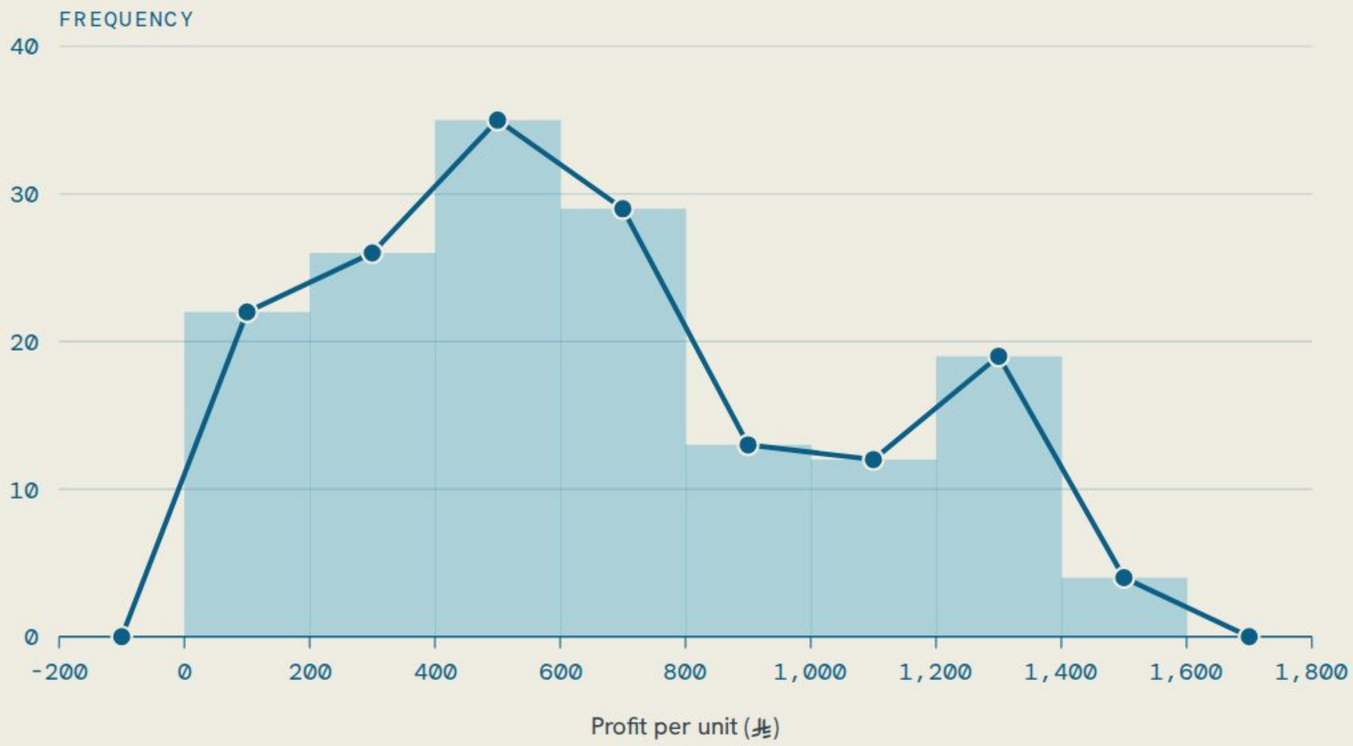


Figure 6 Frequency polygon of the profit on 160 units, drawn over the histogram

The polygon shows the same shape as the histogram, but with line segments instead of bars. The first point is (100, 22): the midpoint 100 has a frequency of 22.

WHY USE IT?

- See the overall pattern of the data at a glance.
- Identify the peak (the class with the highest frequency).
- **Compare** two or more data sets on the same chart; bars would overlap, lines do not.

THINK OF IT AS
Tracing the mountain peaks formed by your data.

Comparing *two* distributions

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

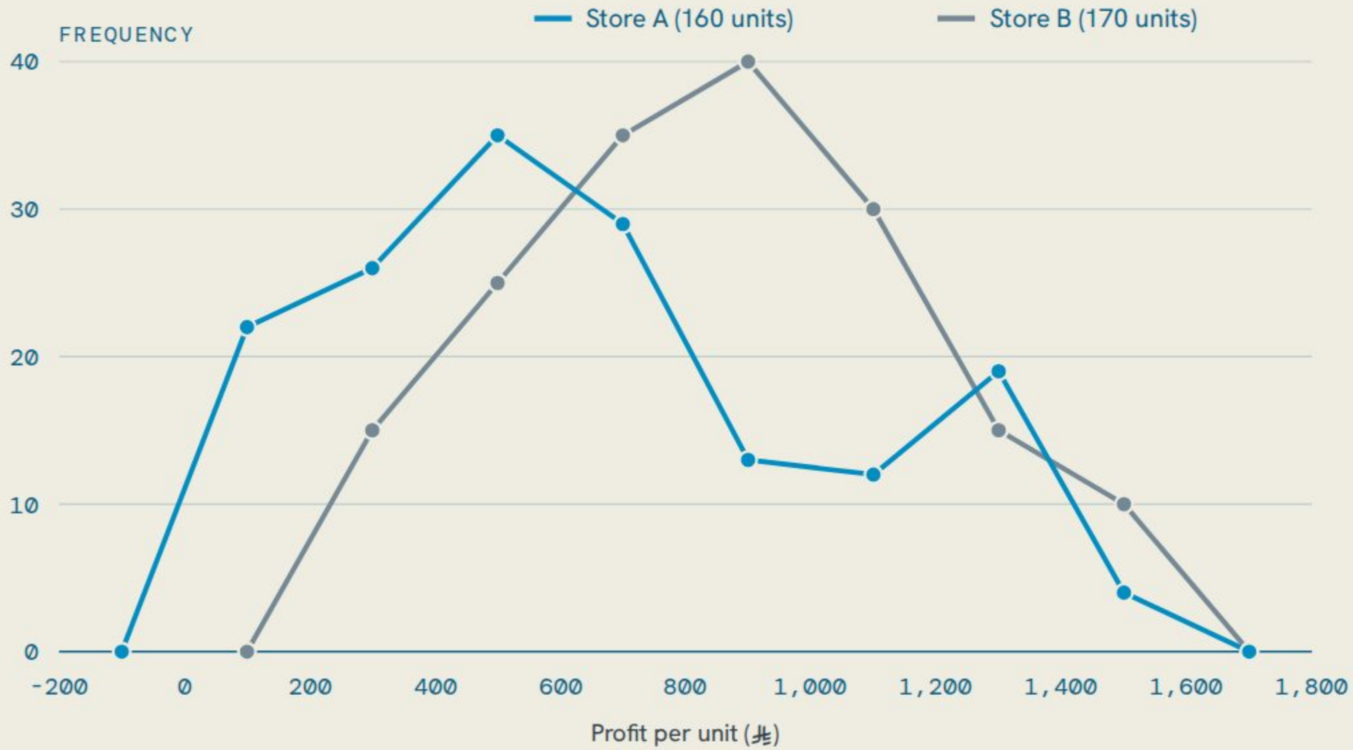


Figure 7 Frequency polygons of the profit per unit at Store A and Store B

READING THE CHART

Store A (blue) is our store: 160 units, peak at 400 up to 600, then a long tail with a second bump. Store B (grey) sold 170 units: its polygon starts later (no profit below ⃀ 200), peaks at 800 up to 1,000, and falls smoothly.

Manager's conclusion: Store B makes more profit per unit; Store A sells many low-profit units.

DIFFERENT GROUP SIZES?

The two stores do not need the same classes or the same number of units. If the sizes are very different (160 vs. 400 units), plot **relative frequencies** (percent) on the vertical axis, so the two lines can be compared fairly.

Cumulative frequency distribution

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

DEFINITION

A **cumulative frequency distribution** shows how many observations are **less than** the upper limit of each class. Start with the frequency distribution and add each frequency to all the frequencies before it.

THINK OF IT AS

Your savings over time: every month adds to the total.

How many units had a profit of less than ⃀ 1,200? And ⃀ 1,200 or more?

Less than ⃀ 1,200: 137. ⃀ 1,200 or more: 160 − 137 = 23.

Table 12 Cumulative frequency distribution of the profit on 160 units

PROFIT	FREQUENCY	CUMULATIVE FREQUENCY	CALCULATION
less than 200	22	22	22
less than 400	26	48	22 + 26
less than 600	35	83	48 + 35
less than 800	29	112	83 + 29
less than 1,000	13	125	112 + 13
less than 1,200	12	137	125 + 12
less than 1,400	19	156	137 + 19
less than 1,600	4	160	156 + 4

Cumulative *relative* frequency

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

Table 13 Cumulative relative frequency distribution of the profit on 160 units

PROFIT	CUMULATIVE FREQUENCY	CUMULATIVE RELATIVE FREQUENCY
less than 200	22	22/160 = 0.1375 (13.75%)
less than 400	48	48/160 = 0.3000 (30.00%)
less than 600	83	83/160 = 0.5188 (51.88%)
less than 800	112	112/160 = 0.7000 (70.00%)
less than 1,000	125	125/160 = 0.7812 (78.12%)
less than 1,200	137	137/160 = 0.8562 (85.62%)
less than 1,400	156	156/160 = 0.9750 (97.50%)
less than 1,600	160	160/160 = 1.0000 (100.00%)

Divide each cumulative frequency by the total (160). The result tells you the **percent** of observations below each class limit, very useful for percentiles and for comparisons.

A MANAGER CAN SAY

- "70% of our units made a profit below ⃁ 800."
- … or the other way round: "30% of our units made a profit of ⃁ 800 or more."
- "97.5% of the units made less than ⃁ 1,400."

Notice the last row: 160/160 = 100%, every observation is below the upper limit of the last class.

The cumulative polygon

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

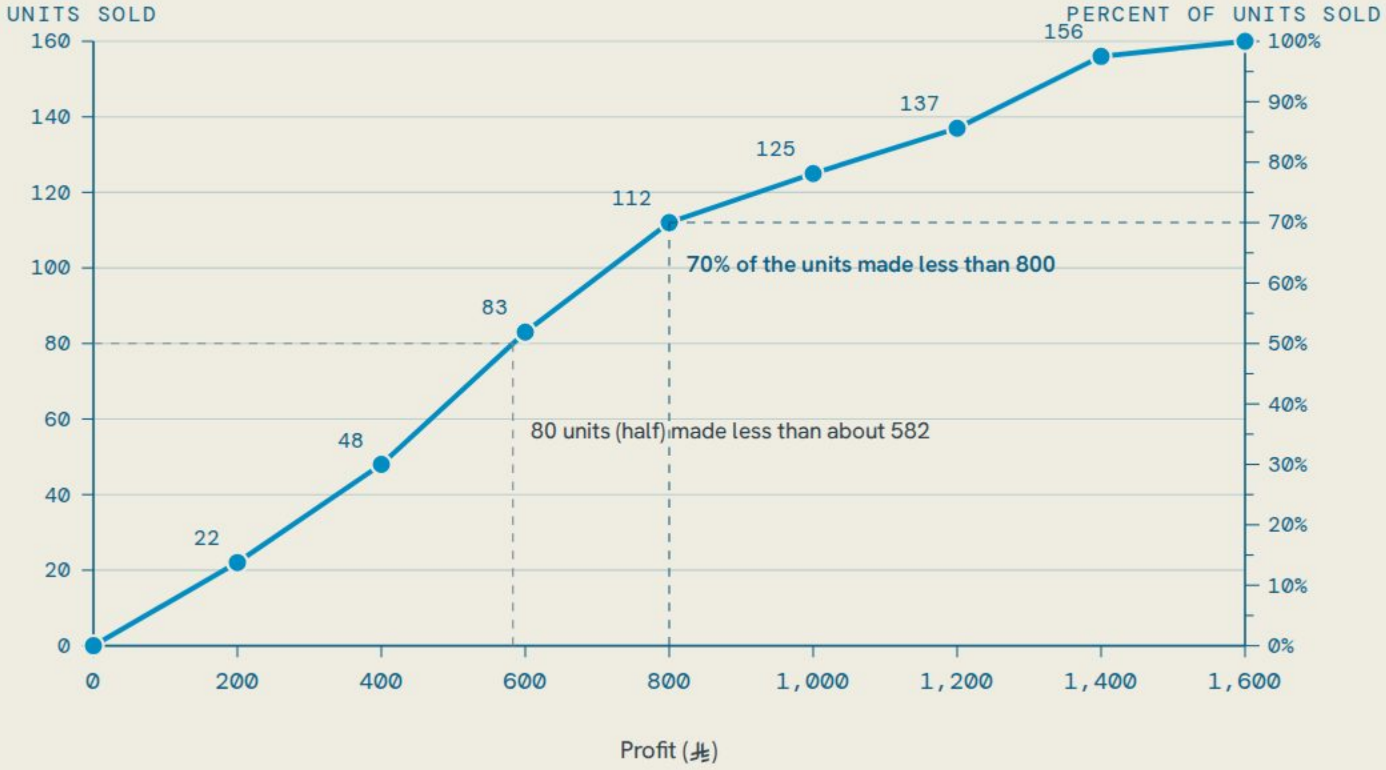


Figure 8 Cumulative frequency polygon of the profit on 160 units

HOW TO DRAW IT

1. Put the **upper limit** of each class on the horizontal axis.
2. Put the cumulative frequency on the vertical axis: number of units on the left, percent of units on the right.
3. Start at (0, 0): nothing is below the lower limit of the first class.
4. Connect the points with straight lines.

SHAPE

It always **rises** from left to right (or stays flat) and ends at the total, 160 = 100%. It never goes down, because a running total cannot shrink.

Reading the cumulative polygon

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

FROM A VALUE TO A PERCENT

What percent of the units made less than ⃁ 800?

Go up from 800 on the horizontal axis to the line, then right to the percent axis: **70%** (that is 112 units on the left axis). The other 30%, 48 units, made ⃁ 800 or more.

FROM A COUNT TO A VALUE

Below which profit are half of the units?

Half of 160 is 80 units. Start at 80 on the left axis, go right to the line, then down: about **⃁ 582**. Half of the units made less than roughly 582 riyals; half made more. (Chapter 3 calls this value the **median**.)

TWO DIRECTIONS

value → count or percent (start on the horizontal axis) · count or percent → value (start on a vertical axis). Same line, read the other way.

Dot plots

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

DEFINITION

A **dot plot** stacks a dot for each observation above its value on a number line. Identical values are stacked on top of each other.

- Shows the shape of the distribution **and** keeps every single value (nothing is grouped into classes).
- Shows where values cluster and where the gaps are.
- Best for **small** data sets, usually fewer than 50 values.

THINK OF IT AS
Coins on a ruler: each coin is a value; you see where the coins pile up.

COMPLAINTS PER DAY · 14 DAYS

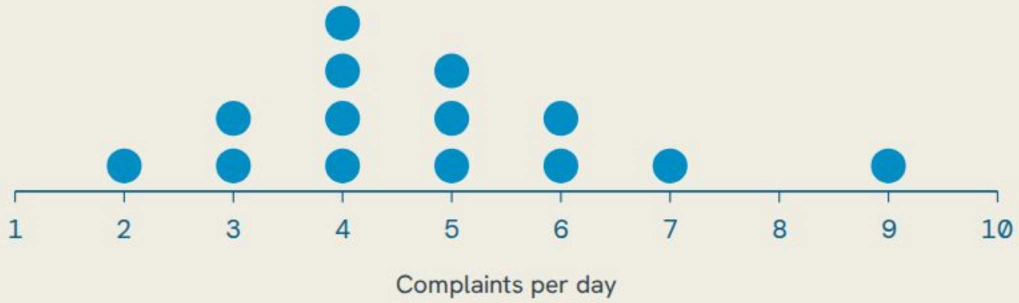


Figure 9 Dot plot of the complaints per day at a call centre, 14 days

Complaints per day at a call centre over 14 days: the most common value is 4 (four days), the values range from 2 to 9, and one day (9) stands apart from the rest.

Example: two Jarir branches

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

Jarir recorded the number of laptops sold per day at two Riyadh branches during the last 30 days. Two small quantitative data sets, perfect for dot plots.

OLAYA BRANCH · 30 DAYS

5	16	19	18	16	9	16	13	14	18
11	7	14	5	16	19	8	11	10	15
16	18	13	9	17	15	19	7	7	13

KHURAIS BRANCH · 30 DAYS

16	17	12	14	15	23	13	22	16	13
14	15	22	14	20	17	11	11	24	16
20	17	17	21	20	19	17	12	21	14

Try before the next slide: what is the lowest and the highest value in each branch? Which value repeats most often? Then look at the dot plots.

Reading the dot plots

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

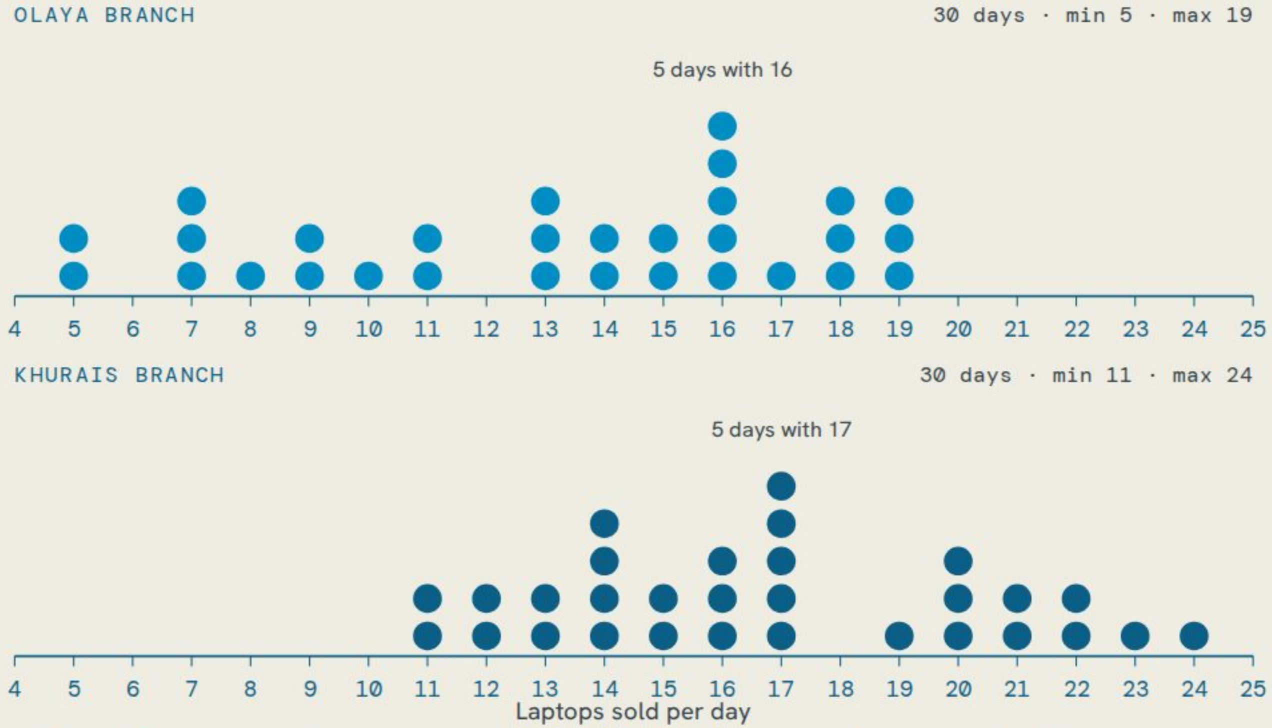


Figure 10 Dot plots of the laptops sold per day at the Olaya and Khurais branches, 30 days each

OLAYA

- Sales per day ranged from 5 to 19 laptops.
- Sales were generally lower and more spread out; a few days with 5 to 9.
- The most frequent value was 16 laptops, on 5 different days.

KHURAIS

- Sold the most laptops in a single day: 24.
- Values cluster between 13 and 17; the most frequent value was 17 (5 days).
- Never fewer than 11 laptops: a higher and steadier branch.

Because the two plots share one scale, you can compare them at a glance: Khurais sits to the right of Olaya.

Check your understanding

LO 2.3 · Frequency polygons, cumulative distributions and dot plots

Q1 What is the midpoint of the class “800 up to 1,000”? Which point of the polygon does it give?

(800 + 1,000)/2 = 900; the point (900, 13).

Q2 From the cumulative table: how many units made less than ⃁ 400? What percent is that?

22 + 26 = 48 units; 48/160 = 30%.

Q3 A cumulative frequency polygon goes down between two points. What is wrong?

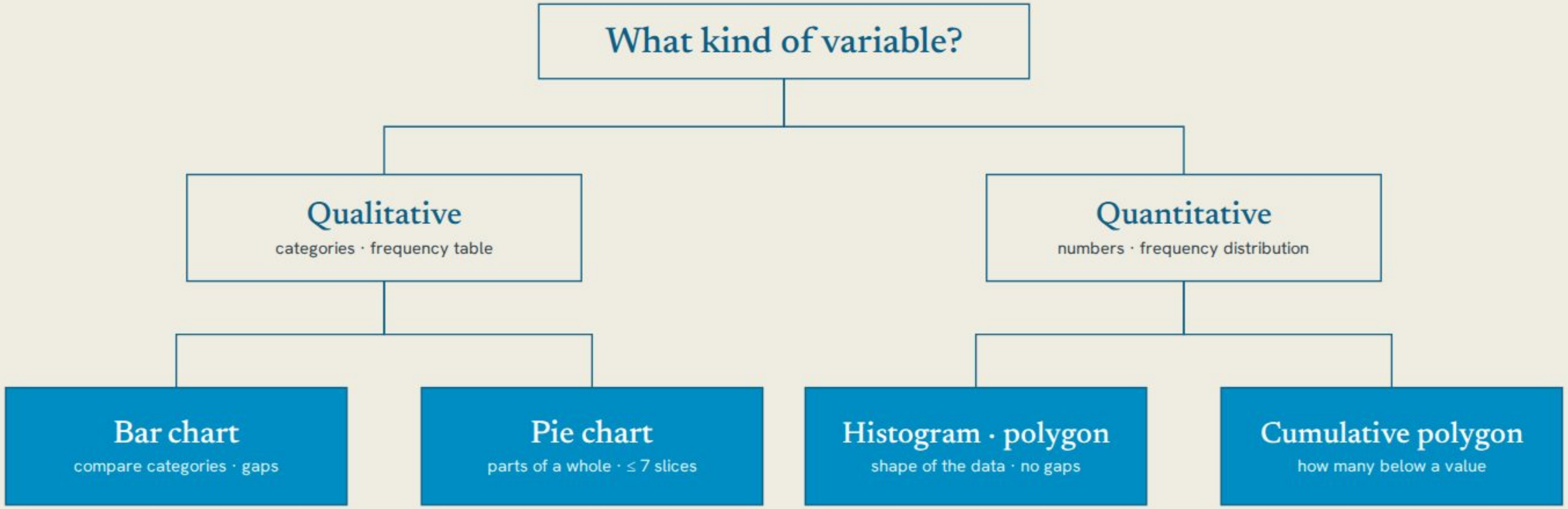
A running total can never decrease, so a frequency was added wrongly (or the points are in the wrong order).

Q4 You have the ages of 18 employees in a small office. Which display keeps every value visible?

A dot plot (small data set, each value shown).

Choosing the right tool

SUMMARY



Small quantitative data set (under 50 values)? A dot plot shows every value.

Figure 11 Choosing the table and the chart by the type of variable and the question

Organize first, draw second. Frequency tables and distributions turn raw data into classes; relative frequencies show shares; cumulative frequencies answer "how many below". The chart depends on the type of data (qualitative or quantitative) and on the question you want to answer.

CHAPTER 2 · PRACTICE

Draw it, then *read it.*

NEXT

Chapter 3: Describing Data: Numerical Measures (mean, median, mode, spread)

← Back to the QUA 107 course page

01 **Chart Builder** ↗

Build the frequency table, then draw the bar chart, histogram, polygon and cumulative polygon yourself.

02 **Chart Detective** ↗

Read charts, pick the right chart for the data, and spot the mistakes.

03 **TheKey practice pools** ↗

CH2LO1 to CH2LO3 · 25 practice pools with many versions of each question.